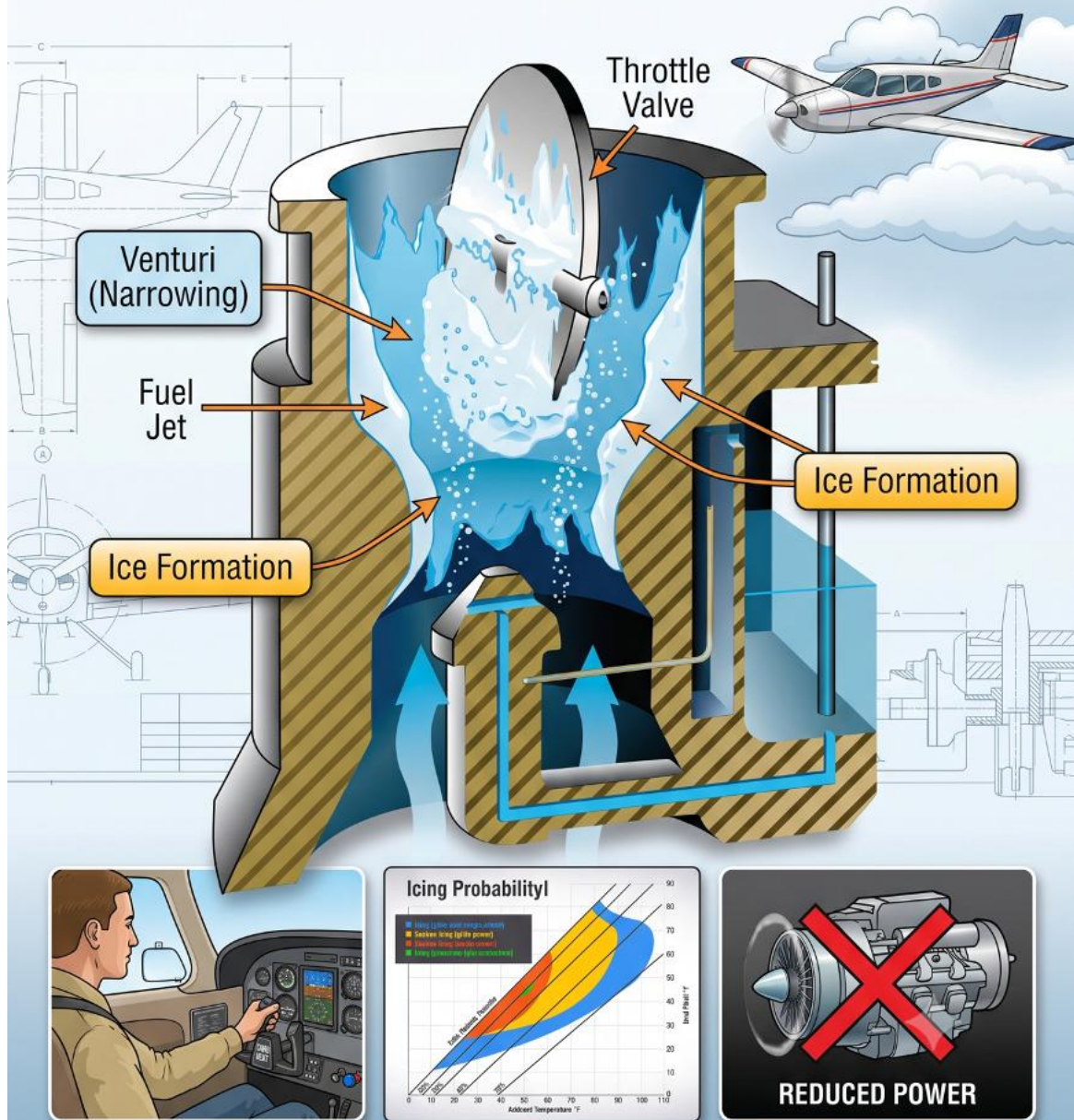


TRAINING MANUAL

CARBURETTOR ICING: DETECT, PREVENT, AND RECOVER



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Basic Pilot Training Manual:

Aircraft Carburettor Icing

Carburettor icing is one of the most common engine icing hazards in light piston-engine aircraft and a critical safety concern for general aviation pilots. It occurs when moisture in the air freezes inside the carburettor, potentially blocking the fuel/air flow to the engine and causing a dangerous loss of power. This phenomenon can even cause engine failure if not recognized and corrected in time. The sneaky part is that carburettor ice can form in conditions that might *not* seem obviously “icy” – even on a warm, humid day. The good news is that carburettor icing can be prevented and quickly resolved if you know what to look for and use the proper procedures. This basic training guide explains how carburettor icing occurs, when it's most likely, how to recognize the warning signs, and what steps to take to prevent and recover from carb icing – all in clear, beginner-friendly terms.

What Is Carburettor Icing?

Definition: Carburettor icing (often called *carb ice*) is ice forming inside the carburettor of a piston aircraft engine. It is caused by freezing of moisture from the air in the carburettor's throat (venturi) or around the throttle valve (butterfly) due to rapid cooling of the incoming air-fuel mixture. The ice buildup restricts the airflow and the fuel/air mixture going into the engine, resulting in a loss of power or even engine stoppage if the ice becomes severe.

How it forms: In a carburetted engine (common on many light Lycoming and Continental engines found in training aircraft), fuel and air mix in a carburettor before reaching the engine cylinders. As air flows through the narrow venturi of the carburettor and around the partially closed throttle plate (especially at low power settings), the air pressure drops and its temperature falls significantly due to the *Venturi effect*. Additionally, evaporation of fuel into the airstream causes further cooling of the surrounding air. This cooling can be dramatic – the temperature inside the carburettor may drop 15–30°C (30–54°F) or more below the outside air temperature. If the air contains water vapor (humidity), this cooling can chill the air/fuel mixture below the freezing point. The moisture then condenses and freezes on the inner surfaces of the carburettor (especially on the throttle valve and venturi walls), forming ice crystals that gradually build up and obstruct the flow of air and fuel.

Example: If you've ever sprayed an aerosol or let fuel evaporate on your skin, you may have noticed a cooling sensation. Similarly, inside a carburettor, a combination of pressure drop and fuel evaporation can rapidly drop the temperature and cause water vapor to turn into ice, even on what seems like a mild day.

Where the ice forms: Carb ice most commonly forms around the throttle (butterfly) valve in the carburettor. This is because the partial blockage of air at low throttle settings creates extra cooling and a prime site for ice formation. Ice can also form in the main venturi throat itself. In either case, the result is the same: reduced airflow and fuel to the engine.

Fuel injection difference: It's important to note that fuel-injected engines are not susceptible to carburettor icing because they do not have a carburettor. However, even fuel-injected engines can experience other forms of induction ice (like impact ice clogging an air filter in freezing rain/snow, which is mitigated by alternate air intakes). For this manual, we focus on carburetted engines, which include many common training aircraft engines (e.g., Lycoming O-320 series in Cessna 172s or Continental O-200# Basic Pilot Training Manual: Aircraft Carburettor Icing (PPL Level)

Carburettor icing (often called *carb ice*) is a common hazard for small piston-engine aircraft. It occurs when moisture in the air freezes inside the carburettor, reducing or blocking the fuel-air flow to the engine. Carb icing can cause significant power loss or even engine failure if not recognized and handled properly, and has been a factor in numerous general aviation accidents. The good news is that carb ice is usually preventable with proper awareness and use of carburettor heat. This manual explains, in simple terms, what carburettor icing is, when and why it happens, how to recognize it, how to prevent it, and how to recover if it occurs. A quick-reference checklist for cockpit use is included at the end.

Carburettor Icing - How Does It Form?

A carburettor mixes air and fuel for the engine. As air flows through the narrow opening (*venturi*) in a carburettor, the air pressure drops and the temperature falls significantly (as much as 15–30°C or 30–60°F cooler than the surrounding air). At the same time, fuel evaporating into the airstream absorbs

heat, causing further cooling. If the incoming air has moisture (water vapor), this rapid cooling can chill the air to below freezing. Carburettor icing is what happens when that moisture freezes inside the carburettor (typically forming ice around the venturi throat or throttle valve (butterfly)). The ice buildup gradually blocks the airflow and fuel mixture passage, starving the engine of power.

Tip:

You can feel the cooling effect of fuel evaporation if you've ever spilled avgas or gasoline on your hand – it feels cold as it evaporates. In the carburettor, that cooling effect can be enough to freeze water vapor into ice!

Where does the ice form? Ice can accumulate in the venturi (narrow throat) and on the throttle plate. In fact, carb ice most often forms on or around a partially closed throttle (butterfly valve) inside the carburettor. This is because at low power settings, the combination of venturi cooling and fuel vaporization is most pronounced, and there's less engine heat to prevent icing. Carb ice is especially likely at low power (idle or reduced throttle), such as during approach or descent. In these situations, a pilot might not notice the ice forming until trying to increase power, at which point the engine may not respond due to ice choking off the intake.

Why is carb icing dangerous? Even a small amount of ice in the carb can reduce engine power. If left unchecked, the ice can continue to accumulate and potentially cause the engine to run very rough or stop due to fuel starvation. Carb ice tends to develop gradually – it's often a *"silent"* hazard until it becomes severe. That's why learning to anticipate and detect carb icing is so important for new pilots.

Note:

Carburettor icing is an issue only in carburetted engines. Many small general aviation (GA) airplanes – including popular trainers with Lycoming or Continental carburetted engines – are susceptible. Fuel-injected engines usually do not get carburettor ice (since they have no carburettor); however, they can still suffer impact icing or other induction icing in extreme conditions (e.g. ice clogging an air intake or filter). Always know whether your aircraft's engine is carburetted (if so – it will have a carb heat control) or fuel-injected (which typically will have an alternate air control for ice emergencies instead).

Year-Round Risk

Not just a cold weather problem:

Carburetor icing can occur on warm, humid days – even when outside air is above 20°C (70°F) – if moisture is high enough. Pilots must stay vigilant even in seemingly mild weather.

Subtle Warning Signs

Clue: Loss of power – The first sign of carb ice is often a **gradual drop in engine RPM** (in fixed-pitch prop planes) or a **drop in manifold pressure** (in constant-speed prop planes) with no other cause. This is usually followed by **engine roughness or vibration** as the mixture becomes disrupted.

Carb Heat is the Cure

Full carb heat = lifesaver: At the first sign of carb icing, **apply full carburetor heat** immediately. The engine may run **rougher at first** – this is normal as melting ice goes through – but will smooth out as the ice clears.

When and Why Carb Icing Occurs – Conducive Conditions

Carburettor icing can occur even when the outside air is well above freezing. In fact, moderate outside temperatures combined with high humidity are the most likely conditions for carb ice.

- **Temperature:** The highest-risk outside air temperature range is roughly +5°C to +20°C (41°F to 70°F), especially when paired with high humidity. However, carb icing *can* occur outside

this range too. It's possible (though less likely) to get carb ice at *colder* temperatures (if there's visible moisture/precipitation) or even on warm days (as high as ~32°C / 90°F) given sufficient humidity. Always remain alert for carb ice if any moisture is present in the air.

- Humidity: High relative humidity (roughly 50% and above) or visible moisture (fog, clouds, rain) greatly ****increases carb ice risk**. The worst-case scenario is very humid air (e.g. >80% RH) at moderate temperature. A useful rule of thumb: if the temperature-dew point spread is narrow (within ~3°C or ~5°F), the air is nearly saturated with moisture – prime conditions for carb ice formation.
- Throttle setting: Low power or partial-throttle operation makes carb icing more likely. At lower power (such as idling, descending, or taxiing), the throttle is partly closed. This increases the venturi effect around the throttle plate, causing additional cooling right where ice tends to form. Also, with the engine producing less heat, the carburettor stays cooler, so ice is less likely to melt on its own. Prolonged glide or taxi with low power can allow significant ice accumulation before any change in engine performance is noticed. Conversely, at high power (high throttle) the engine and intake system run hotter and airflow is more rapid, so carb ice is less common at cruise power – but it *can still occur*, especially if moisture is abundant. In fact, carb ice has caused accidents even during normal cruise flight when pilots didn't expect it.

In summary, the conditions that favour carb ice are moist air together with cooling inside the carburettor. This cooling can happen year-round, not just in winter. Always consider carb icing potential whenever you operate in a humid environment – including mist, rain, clouds, or even high humidity on a clear day – especially if you will be using low or partial power for any period of time.

Recognizing Carb Icing – Signs and Symptoms in Flight

Detecting carburettor icing early is critical. Carb ice usually forms gradually, so the symptoms may sneak up on you. Here are the key signs of carb icing to watch for during flight:

- Loss of RPM or Power: In an aircraft with a fixed-pitch propeller, a gradual drop in engine RPM is the primary warning sign of carb icing. In an aircraft with a constant-speed propeller, rpm is held constant by the governor, so instead you'll see a drop in manifold pressure for no apparent reason. In *either* case, if you notice a slow, uncommented power loss (RPM or manifold pressure decreasing even though you haven't changed the throttle setting), suspect carb icing.
- Engine Roughness: As ice builds up and the fuel-air mixture becomes excessively rich or uneven, the engine may start running rough or vibrating abnormally. You might feel a shudder or hesitation in power delivery. This rough running often comes after the initial power loss if the ice is not cleared promptly.
- Throttle Unresponsive or "Sticky": In advanced cases, adding throttle has little or no effect on power output because the butterfly valve may be stuck or airflow is blocked by ice. The throttle may feel physically stiff or "frozen" (literally) if ice has formed around it, preventing normal movement.
- Other indications: If your aircraft is equipped with an EGT (exhaust gas temperature) gauge, unusually low EGT can be a subtle clue, as a richer mixture (caused by ice restricting airflow) burns cooler. Fuel flow may also increase if the mixture becomes richer due to reduced air intake. However, these indications are secondary – the primary cues are power loss and rough engine.

Remember:

Any unexplained loss of engine power in a carburetted aircraft should raise suspicion of carb icing. It's better to assume icing and apply carb heat *early* than to wait until the engine gets worse. If you catch carb ice early, applying heat can restore power quickly; if you wait too long, the engine could even quit.

Prevention Strategies – How to Avoid Carburettor Icing

The best strategy is to prevent carb ice from forming in the first place. Here are practical measures and operating techniques to minimize the chance of carburettor icing:

- **Pre-Flight Planning:** Check the weather for temperature and humidity. Pay attention to the temperature–dew point spread: a small spread (e.g. dew point within a few degrees of the air temp) suggests high humidity, meaning conditions may favour carb icing. If a humid day is forecast (or flying in visible moisture like fog, cloud, or rain), have carb icing in mind and be ready to use carb heat.
- **Engine Run-Up:** Always test your carburettor heat during pre-takeoff run-up to ensure it's working. Pull the carb heat on and look for the expected slight drop in RPM (typically 50–150 RPM). No RPM drop could indicate a malfunction (e.g., a disconnected cable or air leak), meaning the system might not be providing hot air. If carb heat is *inoperative*, do not fly unless absolutely necessary – a non-working carb heat means you cannot clear ice if it forms.
- **Avoid Unnecessary Low Power in Moist Air:** Limit long periods of idle or very low power when in humid conditions if possible. For example, during flight training manoeuvres or extended descents, consider adding small power increases periodically or using carb heat to keep the engine warm and clear any small ice accumulations (this is sometimes called “clearing the engine”).
- **Use Carb Heat Proactively at Lower Power:** Follow your aircraft's checklist regarding carb heat usage. Many training airplanes (like Cessna 152/172, Piper Cherokee with carburetted engines) recommend using carb heat on every landing approach or whenever reducing power for an extended descent. This ensures warm air is already flowing through the carburettor to prevent ice formation during approach/landing when the engine is at idle or low power. If you're cruising in very humid, cool conditions, you might also apply brief carb heat occasionally (e.g. every 10-15 minutes) to test for any incipient ice – a momentary additional RPM drop followed by a rise could indicate you had some ice that's now melting.
- **Maintain Proper Mixture:** Use the appropriate fuel mixture for the phase of flight. At higher altitudes, lean the mixture during cruise (per your POH). A leaner mixture reduces the amount of fuel evaporating in the carb, which can reduce the cooling effect (and thus reduce the chance of carb ice). Conversely, avoid over-rich mixtures at low power, as excessive fuel can promote more cooling. Always follow recommended procedures for mixture control.
- **Never Use Partial Carb Heat:** Carburettor heat should be either fully on or fully off – never in between. Partial heat is not effective at melting ice and can even worsen icing by introducing warm moist air that then re-freezes in the carburettor. Always apply full carb heat when needed.

By anticipating carb icing conditions and using carb heat proactively, you can often prevent ice from forming to begin with. *Staying ahead of carb icing is far easier than dealing with a serious buildup later on.*

In-Flight Carb Ice Recovery – Using Carb Heat (Step-by-Step)

If you suspect carburettor icing in flight (or confirm it by observing the signs described earlier), take immediate action to restore your engine power. Here is the standard in-flight recovery procedure using carb heat:

1. **Apply Full Carb Heat Immediately:** Pull the carb heat control fully ON without delay. Don't hesitate – the longer you wait, the more ice can form and the harder it will be to melt. Using carb heat directs hot air from the exhaust manifold into the carburettor, raising the temperature inside and beginning to melt the ice.
2. **Maintain Aircraft Control:** Continue flying the airplane (maintain a safe airspeed and glide attitude) while troubleshooting. Aviate first, *then* navigate/communicate.
3. **Expect Initial Roughness:** Be prepared for the engine to run rougher right after you apply carb heat. This happens because the melting ice turns to water and goes through the engine, causing sputtering. Do NOT turn the carb heat off if the engine gets rough – this roughness is actually a good sign that ice is melting and being ingested. Keep the carb heat on.
4. **Monitor Power Recovery:** Watch your RPM or manifold pressure for signs of recovery. As the ice melts and clears, engine power should gradually increase, and the running should become smoother. It may take 30 seconds to several minutes for full power to return, depending on how much ice had accumulated. Be patient and leave carb heat on until the engine is running smoothly at the expected power output.

5. **Adjust Mixture if Needed:** If you're at a high density altitude or you had leaned the mixture, you might need to enrich the mixture slightly after applying carb heat (because warm air is less dense, which effectively riches the mixture). If the engine was already full rich, you typically don't need to change it. Prioritize clearing the ice; mixture can be fine-tuned afterward if necessary.
6. **Assess the Situation:** Once the engine returns to normal power and smooth operation, you can gradually push the carb heat back to cold/off (in stages, to ensure no new ice forms). If power does not recover within 2-3 minutes of applying heat, you should prepare for potential engine failure – declare an emergency and identify a landing site. Extremely severe ice might take a few minutes to clear; but if there's no improvement after several minutes of heat, consider other issues or plan for an immediate landing.

By following these steps, you can usually melt the ice and restore engine power. The key is to act quickly at the first sign of trouble. Delaying carb heat application gives the ice more time to accumulate, which can result in a complete power loss that may not be recoverable in time.

Common Mistakes and Misconceptions about Carb Icing

Even experienced pilots can get tripped up by carb ice if they hold incorrect assumptions or make certain operational errors. Here are some common mistakes and misconceptions to avoid:

- “Carb ice only happens in cold weather.” Wrong! A very common misconception. In reality, carburettor icing often occurs in mild or even warm weather if humidity is high. It's not just a winter problem – in fact, the majority of carb ice incidents happen in temperatures well above freezing.
- “I don't see any moisture, so icing can't happen.” Not true. Carb ice doesn't require visible moisture like rain or fog; it can form in clear air if the air is humid enough (for example, on a humid day with a small temperature–dewpoint spread). Don't let clear skies give you a false sense of security – high humidity is enough.
- Using partial or intermittent carb heat. Avoid this mistake. Carb heat should be used fully (full *ON* or *OFF*). Partial heat can make the problem worse by melting some ice only to have it refreeze, or by raising air temperature just into the icing range without fully clearing the ice. When needed, always use full carb heat until the ice is gone.
- Turning off carb heat too soon due to rough running. When you first apply carb heat during icing, the engine may run extra rough or even lose a few more RPM as melted ice (water) is ingested. Don't mistake this for “making things worse.” A critical mistake is pushing the carb heat back off immediately – many accidents have occurred because pilots did this, thinking they were harming the engine, when in fact they were in the middle of successfully clearing the ice. Always keep carb heat on until the engine performance improves and stabilizes.
- Forgetting to turn off carb heat when needed. While using carb heat is vital to regain power in icing conditions, remember to turn it off when you need full power again (for instance, during a go-around or climb). Leaving carb heat on robs the engine of some power because the heated air is less dense (and often unfiltered). This can impair climb performance. So after successfully clearing ice, restore cold air intake to develop full power (per your POH, usually carb heat *OFF* when full throttle is required).
- Insufficient preflight checks or neglecting carb heat usage procedures. Some carb ice incidents happen simply because the pilot didn't use carb heat when they should have, or failed to check that the system was working. Always follow your aircraft's checklist regarding carb heat, both on run-up and in flight, and don't skip using carb heat in the situations recommended by your instructor or POH (e.g., on base/final approach in many carburetted trainers).

By avoiding these mistakes and respecting the potential for carb icing on every flight, you can greatly reduce the risk of a surprise power loss.

Quick Reference: Carb Icing Checklist

Before Flight / Ground:

- Weather check: Review temperature & humidity. If temp is between about 0°C and 20°C (32–70°F) with high humidity or a small temp-dewpoint spread, plan for possible carb ice.
- Carb heat pre-flight test: During run-up, check carb heat operation (expect a slight RPM drop when applied). Do not depart if carb heat is not functioning properly.

In Flight – Prevention:

- Remain vigilant in humid conditions, even on warm days. *Carb ice can happen at any time moisture and cooling combine.*
- Use carb heat before long idle: When reducing power for descents or landing approach, apply carb heat *before* or when you throttle back. Leave it on during extended low-power operations (e.g. *approach or glide*).
- Clear ice periodically: During prolonged low-power flight (e.g. training manoeuvres or extended descent), intermittently apply carb heat and/or add power for a short time to melt any starting ice. In cruise, consider a brief carb heat application if in high-risk conditions (watch for an RPM change).

In Flight – If Carb Ice Suspected:

1. Carb Heat ON (Full On): *Immediately* apply full carb heat at the first sign of icing (power loss or roughness). Do NOT use partial heat.
2. Maintain Altitude & Airspeed: Keep flying the airplane; be ready for possible further power reduction initially.
3. Expect Rough Running: The engine may run rough as ice melts – keep carb heat on until engine smooths out.
4. Power Restored: Ice should melt and power/rpm increase back to normal over seconds or a minute or two. Turn carb heat off *only* after normal power is restored.
5. If No Improvement: If full heat for ~2-3 minutes does not restore power, prepare for engine failure and possible emergency landing. (Maintain carb heat and try to find warmer air by descending if safe to do so.)

By understanding carburettor icing and following these procedures, student pilots and seasoned pilots alike can safely manage this common hazard. Stay alert to carb icing conditions, use carb heat proactively, and you'll significantly reduce the risk of an unexpected engine power loss due to carburettor ice. Fly safe!

User Manual

How to Use the Carburettor Icing Probability Chart

1. Introduction

This manual explains how to correctly use the **Carburettor Icing Probability Chart**. The chart helps pilots determine the **likelihood and severity of carburettor icing** based on atmospheric conditions such as temperature and moisture.

Carburettor icing can occur even in relatively warm conditions and may affect engine performance. This chart provides a visual method to assess risk before or during flight.

2. Required Inputs

Before using the chart, you must obtain:

- **Outside Air Temperature (°C)**
- **Dew Point Temperature (°C)**

These values are typically found in weather reports.

The chart specifically instructs that the user must first **obtain temperature and dew point** before proceeding.

3. Step 1: Calculate Dew Point Depression

Dew point depression is the difference between:

Temperature – Dew Point = Dew Point Depression

Example:

- Temperature = 12°C
- Dew Point = 2°C
- Dew Point Depression = 10°C

This exact example is provided in the chart instructions.

4. Understanding the Chart Layout

4.1 Horizontal Axis (Bottom Axis)

- Represents **temperature in degrees Celsius**

4.2 Vertical Axis (Left Side)

- Represents **dew point depression**

4.3 Right-Hand Scale

- Indicates **relative humidity reference**

4.4 Shaded Areas

- Represent different levels of **icing probability and severity**

5. Step 2: Locate Your Values

1. Find the **temperature** on the horizontal axis.
2. Find your calculated **dew point depression** on the vertical axis.
3. Draw an imaginary line from each value until they intersect.

The chart instructs users to determine icing probability from **the intersection of these lines**.

6. Step 3: Determine Icing Risk

Once you locate the intersection point, observe the **shaded region** in which it lies.

The chart legend provides categories such as:

- **Serious icing – any power**
- **Serious icing – descent power**
- **Moderate icing – cruise power**
- **Light icing – cruise or descent power**

These categories are defined in the chart's shading legend.

7. Interpreting Results

The shading indicates:

- **Severity of icing**
- **Likely engine power condition where icing occurs**

Important interpretation notes:

- **Descent power** increases susceptibility due to lower engine heat.
- **Higher humidity (low dew point depression)** increases icing probability.
- Icing can occur even in temperatures well above freezing.

All interpretations are based on the chart's labeled shading regions and scales.

8. Example of Full Use

Given:

- Temperature = 20°C
- Dew Point = 15°C

Step 1:

Dew Point Depression = 5°C

Step 2:

Locate 20°C (horizontal) and 5°C (vertical)

Step 3:

Find their intersection

Step 4:

Read the shaded zone → determines icing risk level

(This process directly follows the chart's instructions for use.)

9. Operational Considerations

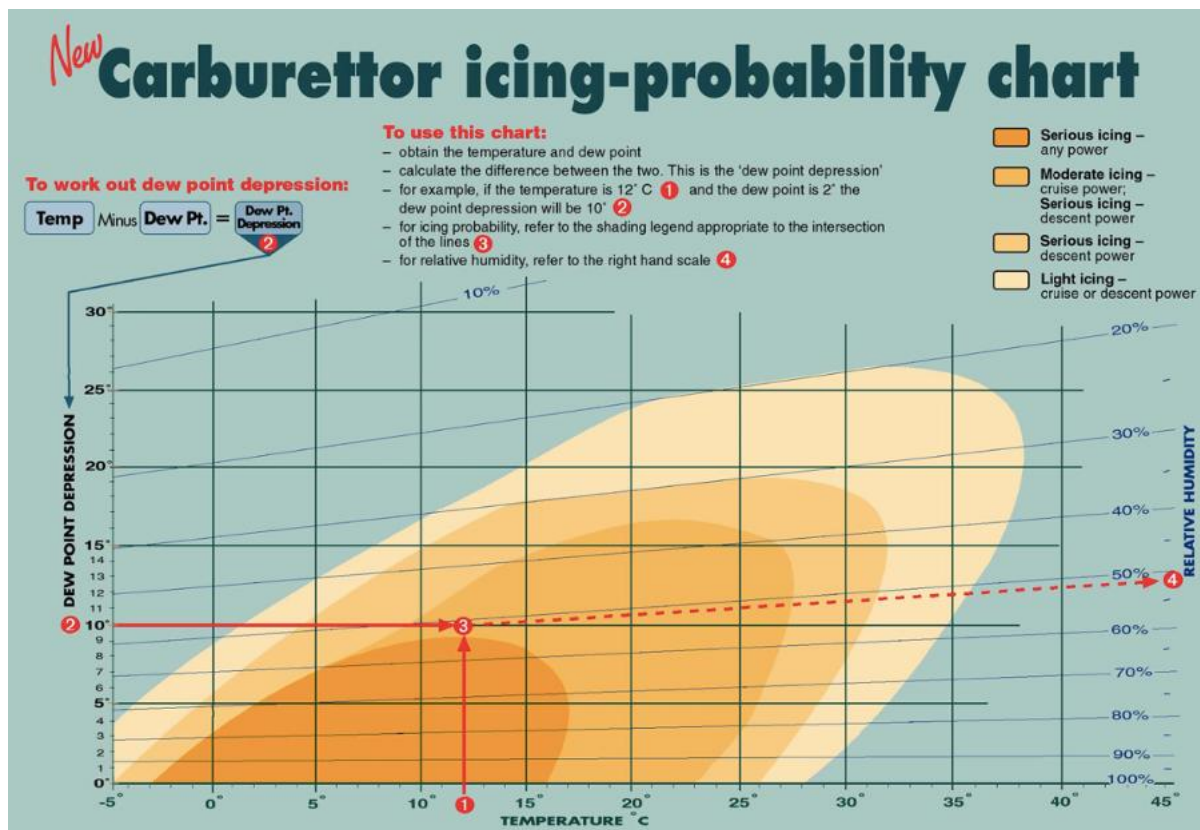
When using the chart in practice:

- Always use **current and accurate weather data**
- Apply results conservatively (assume risk if close to a boundary)
- Monitor engine performance for early signs of icing
- Use carburettor heat where appropriate

(Note: These are practical usage suggestions; the chart itself defines only the method for determining icing probability.)

10. Summary of Procedure

1. Obtain temperature and dew point
 2. Calculate dew point depression
 3. Locate both values on the chart
 4. Find their intersection
 5. Identify the shading zone
 6. Interpret icing probability and severity
-



Carburettor Icing Chart – Quick Reference Card

STEP-BY-STEP USE

1. Obtain:
 - Outside Air Temperature (°C)
 - Dew Point (°C)
2. Calculate:

Dew Point Depression = Temperature – Dew Point
3. Locate:
 - Temperature → bottom axis
 - Dew Point Depression → left axis
4. Find:
 - Intersection point on chart
5. Read:
 - Shaded area = icing risk category

CHART GUIDE

- **Bottom Axis** → Temperature (°C)
- **Left Axis** → Dew Point Depression
- **Right Scale** → Relative Humidity reference
- **Shaded Zones** → Icing probability/severity

ICING RISK CATEGORIES

- **Serious icing – any power**
- **Serious icing – descent power**
- **Moderate icing – cruise power**
- **Light icing – cruise or descent power**

(Determine category from shaded region)

KEY INTERPRETATION TIPS

- Small dew point depression → **higher humidity** → **higher icing risk**

- Icing can occur in **temperatures well above 0°C**
- **Descent power increases risk** due to reduced engine heat
- Use caution near **zone boundaries**

EXAMPLE (MENTAL CHECK)

Temp = 20°C

Dew Point = 15°C

Depression = 5°C

→ Locate 20°C (bottom)

→ Locate 5°C (side)

→ Find intersection

→ Check shaded zone

MEMORY AID

“T-D-D-I-R”

- Temperature
- Dew point
- Difference (depression)
- Intersection
- Risk zone

OPERATIONAL NOTE

Use this chart as a **risk indicator only**. Always monitor engine performance and apply carb heat as required.

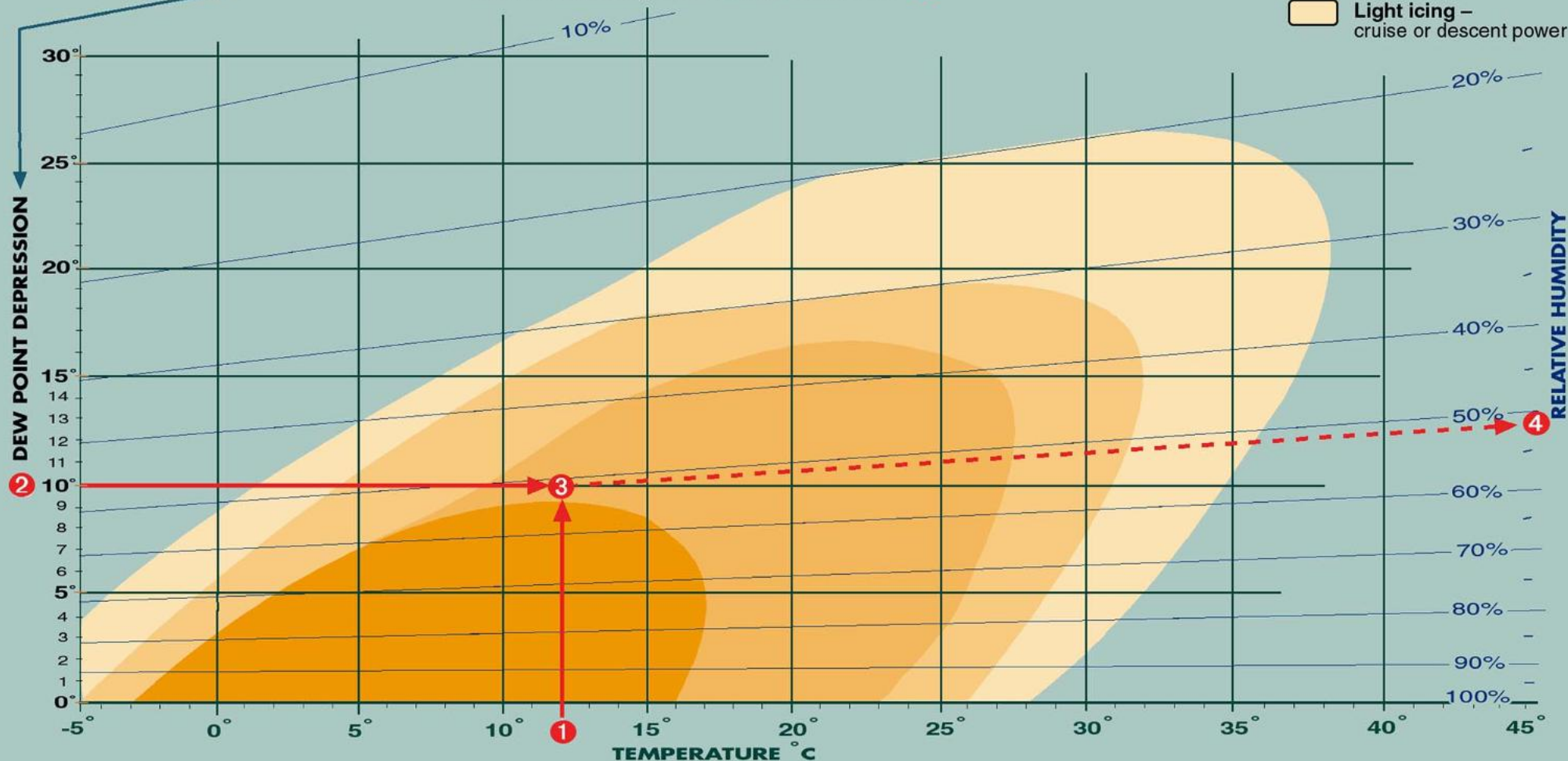
New Carburettor icing-probability chart

To use this chart:

- obtain the temperature and dew point
- calculate the difference between the two. This is the 'dew point depression'
- for example, if the temperature is 12° C **1** and the dew point is 2° the dew point depression will be 10° **2**
- for icing probability, refer to the shading legend appropriate to the intersection of the lines **3**
- for relative humidity, refer to the right hand scale **4**

To work out dew point depression:

$$\text{Temp} \text{ Minus } \text{Dew Pt.} = \text{Dew Pt. Depression}$$



- Serious icing – any power**
- Moderate icing – cruise power; Serious icing – descent power**
- Serious icing – descent power**
- Light icing – cruise or descent power**